

STUDY NOTES — CHAPTER 5

Thorax & Rib Cage

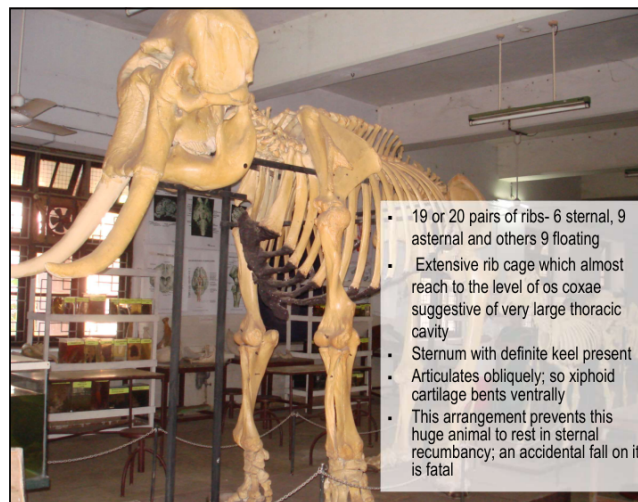
Biological and Anatomical Aspects of Elephants

Dr. Parag Nigam, PhD · Wildlife Institute of India, Dehradun

Training Program: Essentials for Mortality Investigation of Asian Elephant Chhattisgarh Forest Department · 5–6 June 2026, Raigarh | Technical Support: WII Dehradun

CONTENTS: 5.1 Rib Count | 5.2 Species Comparison | 5.3 No Pleural Space | 5.4 Diaphragm | 5.5 Necropsy Findings

5. Thorax & Rib Cage



10

Slide 10 — Thoracic skeleton: extensive rib cage reaching to os coxae level

5.1 Rib Count & Structure

The elephant rib cage is one of the most expansive of any terrestrial mammal, extending posteriorly almost to the level of the os coxae (pelvis). It encloses the enormous thoracic cavity housing a 12–21 kg heart and very large lungs.

Parameter	Asian Elephant	African Elephant
Total rib pairs	19–20	21
Sternal ribs (direct sternum attachment)	6	6
Asternal ribs (costal cartilage)	9	9
Floating ribs (free)	~4–5	~6
Rib cage extent	Reaches nearly to os coxae level	Similar

Thoracic cavity	Very large — accommodates 12–21 kg heart	Very large
-----------------	--	------------

- Sternum: definite keel (carina) present — unlike flat sternum of most domestic animals
- Xiphoid cartilage: articulates obliquely, bends ventrally — relevant to anaesthesia positioning
- Costal cartilages: wide; calcify with age — useful at necropsy for gross age estimation
- Thoracic vertebrae: 19–20 in Asian elephant; spinous processes elongated dorsally

5.2 Comparison with Domestic Species

Species	Rib Pairs	Pleural Space	Breathing Mechanism
Asian Elephant	19–20	ABSENT — pleura fused	Diaphragm ONLY
African Elephant	21	ABSENT — pleura fused	Diaphragm ONLY
Horse	18	Present — fluid-filled	Diaphragm + rib cage expansion
Cattle	13	Present — fluid-filled	Diaphragm + rib cage expansion
Dog	13	Present — fluid-filled	Diaphragm + rib cage expansion
Human	12	Present — fluid-filled	Diaphragm + rib cage expansion

5.3 No Pleural Space — Unique Respiratory Anatomy

Elephants are the **ONLY** mammals with no true pleural space. Both visceral and parietal pleura are composed of dense connective tissue and are **FUSED** together via loose connective tissue. Lungs are physically adhered to the chest wall and cannot collapse passively. Breathing is driven **ENTIRELY** by diaphragm movement.

- Both pleural layers fused — no fluid-filled cavity between them
- Lungs physically adhered to chest wall — cannot collapse passively as in other mammals
- Diaphragm unusually thick and muscular — generates **ALL** inspiratory pressure alone
- Resting respiratory rate: ~4–8 breaths/min; large tidal volume compensates for slow rate
- Evolutionary reason: semi-aquatic ancestry — swimming elephants breathe via trunk-snorkel; pleural fusion prevents lung compression at depth
- West JB (2001): confirmed biophysical significance for deep-water diving/swimming

5.4 Diaphragm — Structure & Function

- **PRIMARY** and near-exclusive respiratory muscle in elephants
- On contraction: pulls lungs caudoventrally → increases thoracic volume → inspiration
- On relaxation: thoracic volume decreases → passive expiration
- Intercostal muscles play **MINIMAL** role compared to other species
- Central tendon anchored to caudal sternum border and thoracic vertebral column
- Diaphragm weight ~2–4% of total body weight — reflects enormous musculature required

CRITICAL DANGERS: 1. STERNAL RECUMBENCY = FATAL — abdominal organs crush diaphragm → respiratory arrest within minutes. ALWAYS lateral recumbency (left lateral preferred). 2. NO THORACOCENTESIS — no pleural cavity to drain; fluid in connective tissue is a clinical emergency. 3. ANAESTHESIA: IPPV parameters differ from other large mammals — standard equine/bovine protocols NOT applicable. 4. NECROPSY: Firm lung-chest adhesion is NORMAL — do NOT diagnose as fibrinous pleuritis.

5.5 Necropsy Findings — Thorax

- Lungs appear **FIRMLY ATTACHED** to inner chest wall — **NORMAL** anatomy, not adhesion from pleuropneumonia
- Any fluid between chest wall and lung surface is **PATHOLOGICAL** (haemorrhage, exudate, oedema)
- Heart: centrally located; **BIFID** (double-pointed) apex — distinctive and normal; weight 12–21 kg
- Lungs: large, soft, pink-grey; spongy texture; mild congestion common post-mortem (gravitational)
- Trachea: large-diameter; check for foreign bodies, mucus plugs, parasites (Mammomonogamus)
- Pericardium: check for haemopericardium, fibrinous deposits (EEHV in calves)

- Intercostal muscles: haemorrhage may indicate electrocution (lightning or poaching)
- Diaphragm: inspect for tears, herniation, parasitic cysts (*Taenia* spp.) on peritoneal surface

MORTALITY INVESTIGATION — THORAX CHECKLIST: • Measure heart weight; document bifid apex; check for EEHV haemorrhages (calves) • Describe lung colour, texture, consolidation, parasites, foreign bodies • Document any fluid between lung and chest wall (volume, colour, odour, consistency) • Collect tracheal swabs for respiratory pathogen PCR (EEHV, *Mycobacterium*) • Inspect diaphragm integrity — tears common in HEC fatalities • Intercostal muscle haemorrhage: photograph and sample for histology (lightning/electrocution)